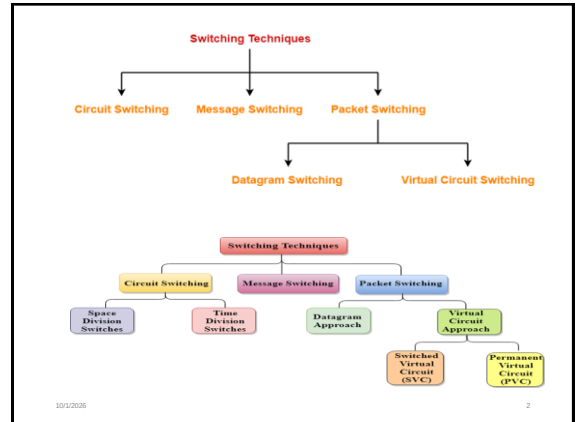


Switching

- Information in a network must move from end-point to end-point switching is required to connect end points
- Data routed by being switched from node to node
- Long distance transmission is done over a network of switched nodes
- Node to node links usually multiplexed
- End devices are called stations:
 - Computer, terminal, phone, etc.

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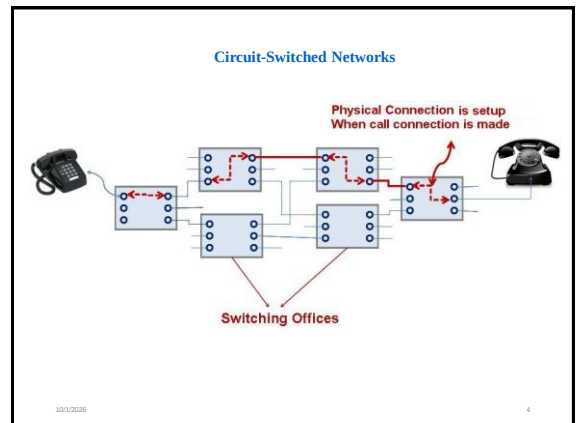
2

Circuit-Switched Networks

- Each session is allocated a fixed fraction of the capacity on each link along its path
 - Dedicated resources
 - Fixed path
 - If capacity is used, calls are blocked
 - E.g., telephone network
 - Advantages of circuit switching
 - Fixed delays
 - Guaranteed continuous delivery
 - Disadvantages
 - Circuits are not used when session is idle
 - Inefficient for bursty traffic
 - Circuit switching usually done using a fixed rate stream (e.g., 64 Kbps)
- Difficult to support variable data rates

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Circuit-Switched Networks

Circuit-Switched networks refers to fact that there is a dedicated communication path between the two stations.

Communication via circuit-switching involves

- Circuit Establishment
 - Point to Point from terminal node to network
 - Internal Switching and multiplexing among switching nodes.
- Data Transfer.
- Circuit Disconnect.

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Advantages

Once connection is established

Network is transparent.
Nodes seems to be directly connected.
Fixed data rate with no delay.

Drawbacks

Can be inefficient

Resources are dedicated to connection even if no data is sent.

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Problems with circuit switching

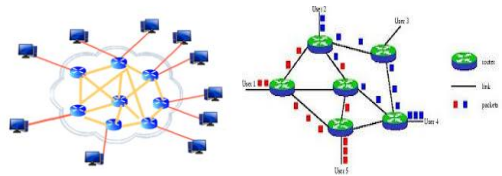
- The rate allocated to the session must be large enough to meet the delay requirement. This allocated capacity is idle when the session has nothing to send.
- If communication is expensive, then circuit switching is uneconomic to meet the delay requirements of bursty traffic
- Also, circuit switching requires a call set-up during which resources are not utilized. If messages are much shorter than the call set-up time then circuit switching is not economical (or even practical)
 - More of a problem in high-speed networks

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Store-and-Forward Switching

- Data transmitted at full link capacity, but links can be shared by multiple sessions on a demand basis.



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Store-and-Forward Switching

- Advantages:
 - Link fully utilized if there is any data to transmit
 - Delay can be significantly reduced
 - Utilization can be significantly increased
- Disadvantages:
 - Greater variance in delay due to queuing delays
 - Flow control needed to prevent buffer overflows

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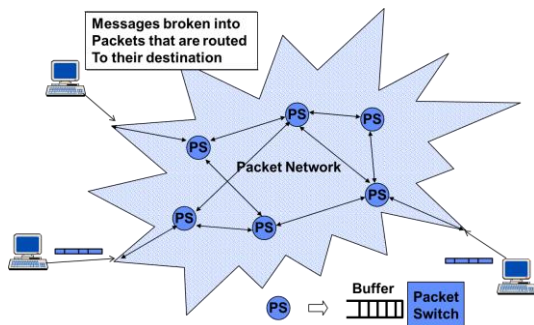
Store-and-Forward Switching

- How is information switched?
 - *Message Switching*: messages are sent intact without being broken into packets
 - *Packet Switching*: messages are broken into packets for transmission
 - *Cell Switching*: messages (or packets) are broken into fixed-size packets called *cells*

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Messages broken into
Packets that are routed
To their destination



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Message switching

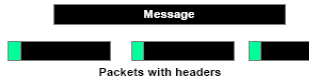
- Networks may be classified by how they segment data for transmission
- *Message-switched* versus *packet-switched*
- Most networks use packet switching (or cell switching)
- Complete message has to be received before sent out again.
- messages are sent intact without being broken into packets

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Message versus Packet Switching

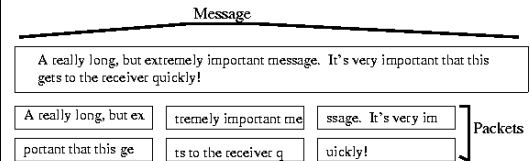
- Packets
- Messages may be decomposed into one or more packets for transmission, reconstructed at receiver
- Lower layer entities may further decompose packets,
- for example: Ethernet frames, ATM cells



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MESSAGE SWITCHING BECOMES PACKET SWITCHING IF "LOGICAL" MESSAGES ARE BROKEN UP INTO SMALLER UNITS (CALLED *PACKETS* WHOSE SIZES ARE CONVENIENT FOR THE NETWORK BEFORE TRANSMISSION.



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Message vs Packet Switching

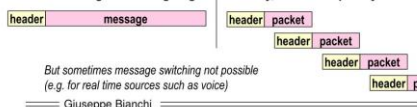
→ Message Switching

- ⇒ One single datagram
- ⇒ Either received or lost
- ⇒ One single network path

→ Packet Switching

- ⇒ Many packets generated by a same node and belonging to a same destination
- ⇒ may take different paths (and packets received out of order -- need sequence)
- ⇒ May lose/corrupt a subset (what happens on the message consistency?)

Message switching: higher reliability, lower complexity



Giuseppe Bianchi

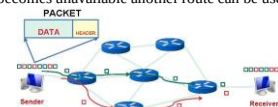
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Packet-Switching Advantages

Packet-switching technique offers:

- Higher utilization efficiency
 - links can be dynamically shared.
- Support of multiple data rates for different types of end points
 - each one connects to a node at proper rate
- No blocking option when load on the network is high, but high delivery time.
- Robustness if one route becomes unavailable another route can be used.

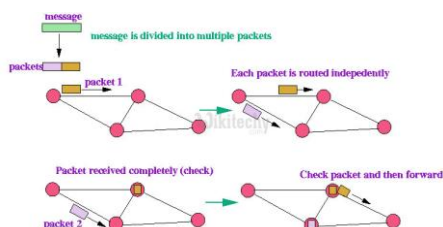


- Switching can be made via two techniques [Datagram](#) and [Virtual Circuit](#)

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Packet-Switching Advantages



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- Advantages of packet switching
 - Efficient for bursty data
 - Easy to provide bandwidth on demand with variable rates
- Disadvantages of packet switching
 - Variable delays
 - Difficult to provide QoS assurances (Best-effort service)
 - Packets can arrive out-of-order

Switching Technique

Circuit switching
Packet switching
Virtual circuits
Datagram

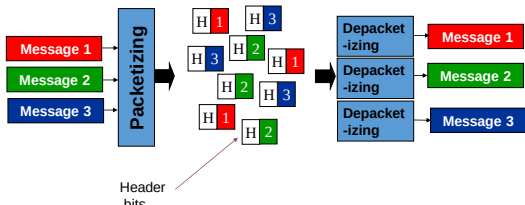
Network service

=> Synchronous (e.g., voice)
=> Asynchronous (e.g., Data)
=> Connection oriented
=> Connectionless

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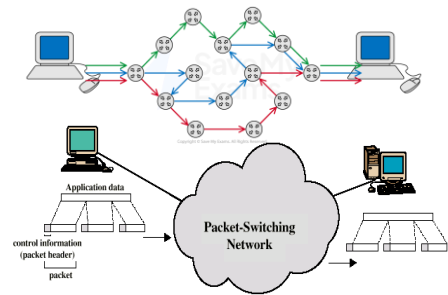
Packet Switching



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Packet-switched networks



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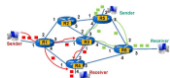
Virtual Circuit Packet Switching Networks

vs.

Datagram Packet Switching Networks

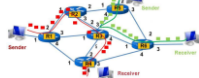
•Datagram:

- Each packet is treated independently.
- Each packet has a full address of the destination.
- Routing decision is taken for each packet at each node.
- Different packets of one message may take different routes.



•Virtual Circuit:

- A connection is setup prior to data transfer.
- Each packet contains a VC identifier.
- Routing decision is made once for all packets.
- All packets follow the same route.



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Circuit Switching Vs Packet Switching

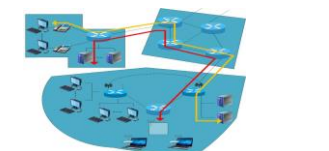
Circuit Switching	Packet Switching
Physical path between source and destination	No physical path
All packets use same path	Packets travel independently
Reserve the entire bandwidth in advance	Does not reserve
Bandwidth Wastage	No Bandwidth wastage
No store and forward transmission	Supports store and forward transmission

3

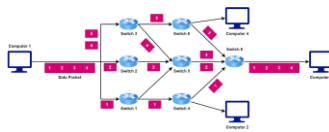
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Virtual Circuit Packet Switching Networks



Datagram Packet Switching Networks

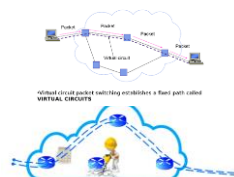


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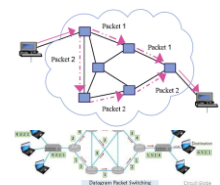
Virtual Circuit Vs Datagram Packet Switching

VIRTUAL CIRCUIT PACKET SWITCHING



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DATAGRAM PACKET SWITCHING



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Why Packetize

- For end to end route comprising of many links, packetizing allows for parts of message to be received, processed, and forwarded while others are still being prepared.
- Amount of retransmitted data due to errors is reduced.
- Memory capacity of internal network nodes can be reduced.
- Transmission time can be reduced.

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How are messages or packets routed through the network?

- *Virtual Circuit Routing*: a path is established and used for the duration of the session.
- *Dynamic Routing*: each packet or message may traverse a different path through the network.

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